The maximum number of stable matchings of order six is forty-eight

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Abstract

Let $f(n)$ denote the maximum number of stable matchings of a stable-marriage instance with n men and n women (Knuth 1976; Gusfield–Irving Open Problem #1). The known values were $f(1), \ldots, f(5) = 1, 2, 3, 10, 16$; the value of $f(6)$ was open, with the lower bound 48 (a dihedral Latin instance) conjectured exact (OEIS A357271). We prove $f(6) = 48$. The proof is a reduction of the $\sim 10^{28}$-instance search space to a finite space of *rotation schedules*: sequences of cyclic partner exchanges applied from the identity matching. Three elementary lemmas over the standard rotation theory show that every instance is dominated, in stable-matching count, by the *read-off instance* of its own schedule, and that a first-appearance labeling and a commutation normal form lose no schedules. An exhaustive enumeration then evaluates the exact stable-matching count of all 26,574,282,886 schedule nodes — roughly ten hours on a laptop — attaining maximum 48. The methodology is validated against ground truth: run at orders $3, 4, 5$ it reproduces $3, 10, 16$, the last being the value the author's companion development certifies formally (Lean + verified SAT certificates); an independent implementation and canonicalization on/off comparisons (up to a 2.66×10^9-node check) agree throughout.

1 Introduction

A stable-marriage instance of order n equips n men and n women each with a strict preference order over the opposite side; a perfect matching is *stable* if no man and woman prefer each other to their assigned partners. Knuth [1] asked for the behaviour of $f(n)$, the maximum number of stable matchings over instances of order n; Gusfield and Irving made it Open Problem #1 of their monograph [2]. Asymptotically $2.28^n \leq f(n) \leq 3.55^n$ [3, 4, 5]. Exact values have been scarce: $f(4) = 10$ and $f(5) = 16$ are due to Eilers [6] (the latter obtained in 2022 by constraint programming and certified formally in the author's companion work [7]), and

$$f(1), \ldots, f(5) \;=\; 1,\; 2,\; 3,\; 10,\; 16.$$

For $n = 6$ the best lower bound has long been 48, attained by a Latin instance built from the dihedral group; OEIS A357271 records the conjecture that 48 is exact. The instance space of order six ($\sim 10^{28}$ after obvious symmetries) is far beyond direct search, and the best human-provable upper bound, $3.55^6 \approx 2005$, is forty times too large.

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Result and method. We prove $f(6) = 48$. The key is a change of universe: instead of instances, we enumerate *schedules* — sequences of cyclic partner exchanges ("rotations") applied from the identity matching, subject to the budget constraints that rotation theory imposes (at most 30 moved-man slots, at most 5 moves per man, no partner revisited). Each schedule *reads off* an instance whose preference lists are the participants' partner trajectories, completed at the bottom. Three lemmas (Section 3) show: (i) every instance's rotations, in elimination order, form a valid schedule; (ii) the read-off instance of that schedule has at least as many stable matchings (the *bridge lemma*); and (iii) a first-appearance labeling rule and a commutation normal form preserve at least one representative of every schedule, without affecting read-off counts. The schedule space is small enough to exhaust: our enumerator visits 26,574,282,886 nodes and computes the exact stable-matching count of every node's read-off instance, attaining maximum 48 (Section 4). With the dihedral lower bound, $f(6) = 48$ follows.

Validation. Because the result rests on an enumeration, Section 5 documents the checks: the same machine run at orders $3, 4, 5$ reproduces $f = 3, 10, 16$ — the last value being formally certified (machine-checked in Lean, with SAT refutations verified by the formally verified checker cake_lpr) in the companion work [7], so a certified theorem serves as test oracle for the present method; an independently written enumerator agrees on maxima; disabling the symmetry reductions multiplies the tree by up to $48\times$ (a 2.66×10^9-node check at order six) and never changes a maximum; and 1.3×10^9 local-search evaluations in two other search spaces also never exceeded 48.

2 Preliminaries

We use the structure theory of the stable-matching lattice and its rotations (Gusfield–Irving [2], Chapters 2–3) and record what we need.

Fact 1. *Call (m, w) a* stable pair *of an instance I if some stable matching contains it. For each man m, his stable partners ordered by his preference form his* trajectory *T_m: the man-optimal matching μ_0 gives each man the first entry of his trajectory, and the rotations of I, applied from μ_0 in any linear extension of the rotation poset, move each man strictly down his trajectory, one entry per rotation containing him, ending at the woman-optimal matching. Symmetrically each woman moves strictly up her trajectory. A rotation is a cyclic sequence $(m_0, w_0), \ldots, (m_{k-1}, w_{k-1})$, $k \geq 2$, of pairs of the current matching, whose application gives m_i the partner w_{i+1} of m_{i+1} (indices mod k). The number of stable matchings equals the number of downsets of the rotation poset.*

Fact 2. *Each man is moved by exactly $|T_m| - 1 \leq n - 1$ rotations, so the total move count satisfies $\sum_\rho |\rho| = \sum_m (|T_m| - 1) \leq n(n-1)$; for $n = 6$: at most 30 slots, at most 5 moves per man, and every rotation moves at least two men.*

3 The schedule reduction

Definition 1 (Schedule)**.** *Fix $n = 6$ and start from the identity matching. A* schedule *is a sequence of* steps*, each a cyclic move on $k \geq 2$ men (man m_i takes the current wife of m_{i+1}), subject to: total move count ≤ 30, at most 5 moves per man, and no man ever moving to a wife he has already had, nor any woman receiving a husband she has already had.*

Definition 2 (Read-off instance)**.** *Given a schedule S, the read-off instance $R(S)$ ranks, for each man, his trajectory in order at the top of his list, followed by the remaining women in a fixed canonical order; for each woman, her trajectory* reversed *at the top, followed by the remaining men canonically.*

Lemma 1 (Validity)**.** *For any instance I of order 6, relabel women so that the man-optimal matching is the identity. Then the rotations of I, in any linear extension of the rotation poset, form a schedule $S(I)$.*

Proof. Rotations are cyclic moves on $k \geq 2$ men exposed in the current matching, and applying them in a linear extension realizes the lattice traversal, in which men move strictly down and women strictly up their trajectories (hence no repeats). The caps are Fact 2. □

Lemma 2 (Bridge)**.** *For every instance I of order 6 (women relabelled as above), $\mathrm{sc}(I) \leq \mathrm{sc}(R(S(I)))$.*

Proof. Write $J = R(S(I))$. The trajectories of $S(I)$ are the stable-partner sequences of I (Fact 1); thus for each man, J ranks his I-stable partners in the same relative order as I does, with all other women strictly below, and symmetrically for women. We claim every I-stable matching μ is J-stable, which proves the inequality. Every pair of μ is a stable pair, hence a trajectory pair on both sides. Let (m, w), $w \neq \mu(m)$. If $w \notin T_m$: in J, m ranks w below all of T_m, in particular below $\mu(m)$; no block. If $w \in T_m$: then (m, w) is a stable pair, so $m \in T_w$; also $\mu(m) \in T_m$ and $\mu^{-1}(w) \in T_w$. Since J preserves relative I-order within trajectories on both sides, (m, w) blocks μ in J iff it blocks in I — and it does not. □

Lemma 3 (Symmetry soundness)**.** *(i) Exchanging two adjacent steps with disjoint man-sets leaves all trajectories, hence $R(S)$, unchanged. (ii) Relabelling men and women by a common permutation maps schedules to schedules and read-off instances to relabelled instances of equal count. Consequently an enumeration that (a) requires the new men of each step to be exactly the next unused labels as a set, and (b) rejects a step lex-smaller than an earlier step separated from it only by man-disjoint steps, still attains* $\max_S \mathrm{sc}(R(S))$.

Proof. (i) Steps with disjoint man-sets touch disjoint women (their women are their men's current wives). (ii) Immediate. For the consequence, relabel by first participation (any assignment within a step's new set satisfies the set rule), and note rule (b) discards a sequence only in favour of a commutation-equivalent one with the same read-off instance. □

4 The enumeration

The enumerator (`gen_enum.c`, ∼200 lines of C) performs a depth-first search over schedules: steps are the 409 cyclic moves on 2..6 men in min-first form; state comprises the current matching, per-person trajectories and visit masks; the reductions of Lemma 3(a,b) and the caps of Definition 1 prune the tree. At *every* node the read-off instance is built (trajectory prefix + bottom completion) and its stable-matching count computed exactly by scanning all 720 perfect matchings. Sharding on the depth-1 step splits the work across cores.

The full run (512 shards, 8 cores of an Apple M4 Max, ∼10 hours) visits 26,574,282,886 nodes, evaluates all of them, and attains

$$\max_S \ \mathrm{sc}(R(S)) \ = \ 48.$$

Because every node is evaluated, no monotonicity assumption about schedule extension is needed: for any instance I, the node $S(I)$ itself is evaluated (Lemma 3).

Theorem 1. $f(6) = 48$.

Proof. The dihedral Latin instance attains 48 (direct computation, three independent counters; OEIS A351413). Conversely, for any instance I, $\mathrm{sc}(I) \le \mathrm{sc}(R(S(I))) \le 48$ by Lemmas 1–3 and the enumeration. □

5 Validation

Ground truth. The same enumerator, parametrized by order, reproduces every known value: order 3 (8 nodes, max 3), order 4 (409 nodes, max 10), order 5 (498,599 nodes, max 16). The last is decisive: $f(5) = 16$ is certified in the companion development [7] by a Lean 4 proof with SAT refutations checked end-to-end by the formally verified checker cake_lpr — a machine-checked theorem serving as test oracle for the present method.

Symmetry on/off. Disabling both reductions of Lemma 3 multiplies the tree (4.5× at order 4; 14× at order 5; 48× in a budget-16 order-6 check of 2,657,773,436 nodes) and never changes a maximum.

Independent implementation. A separately written Python enumerator (different pruning, different code path) agrees on the maxima at the budgets where it is feasible.

Other search spaces. Before the enumeration, 9.7×10^7 hill-climbing evaluations in raw instance space and 1.2×10^9 evaluations in swap-schedule space (72 million valid schedules) never exceeded 48; an earlier maximal-only pass of the full enumeration (1.40×10^{10} exact counts) also attained 48.

Structural corroboration. All 450 sampled order-5 maximizers share a single rotation-poset skeleton; the dihedral order-6 instance uses the full budget of fifteen size-2 rotations; and across the enumeration only the shard containing the dihedral family reaches 48, with large-rotation shards peaking strictly lower — consistent with extremal instances being full-budget, all-size-2 phenomena.

6 Related work and outlook

The computational lineage on $f(n)$ is Eilers' ($f(4)$, $f(5)$, and the extremal-profile counts, which we replicated independently in [7]), building on Thurber [3] and Knuth [1]; the asymptotic bounds are [4, 5]. Certified computational values of comparable flavour include $R(4, 5) = 25$, the empty hexagon number, Keller's conjecture and Schur number five; our companion paper [7] follows that tradition for $f(5)$, and certifying the present enumeration (a verified replay of the DFS, and the bridge lemma in Lean — an elementary argument over finite lists) is natural future work. The next open value, $f(7)$ (best lower bound 71, budget 42), appears to need either a much faster certified evaluator or new structural cuts: the schedule tree grows by roughly three orders of magnitude.

Artifact. All code, logs and verification instructions accompany the repository; the enumeration is reproducible in ∼10 laptop-hours.

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References

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